ANDROID APPLICATION DEVELOPMENT LAB

MCA II B (1)

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25/SCA/MCAN/051

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Program 1: Create an application that takes the name from a text box and shows hello message along with the name entered in text box, when the user clicks the OK button.

activity_main.xml:

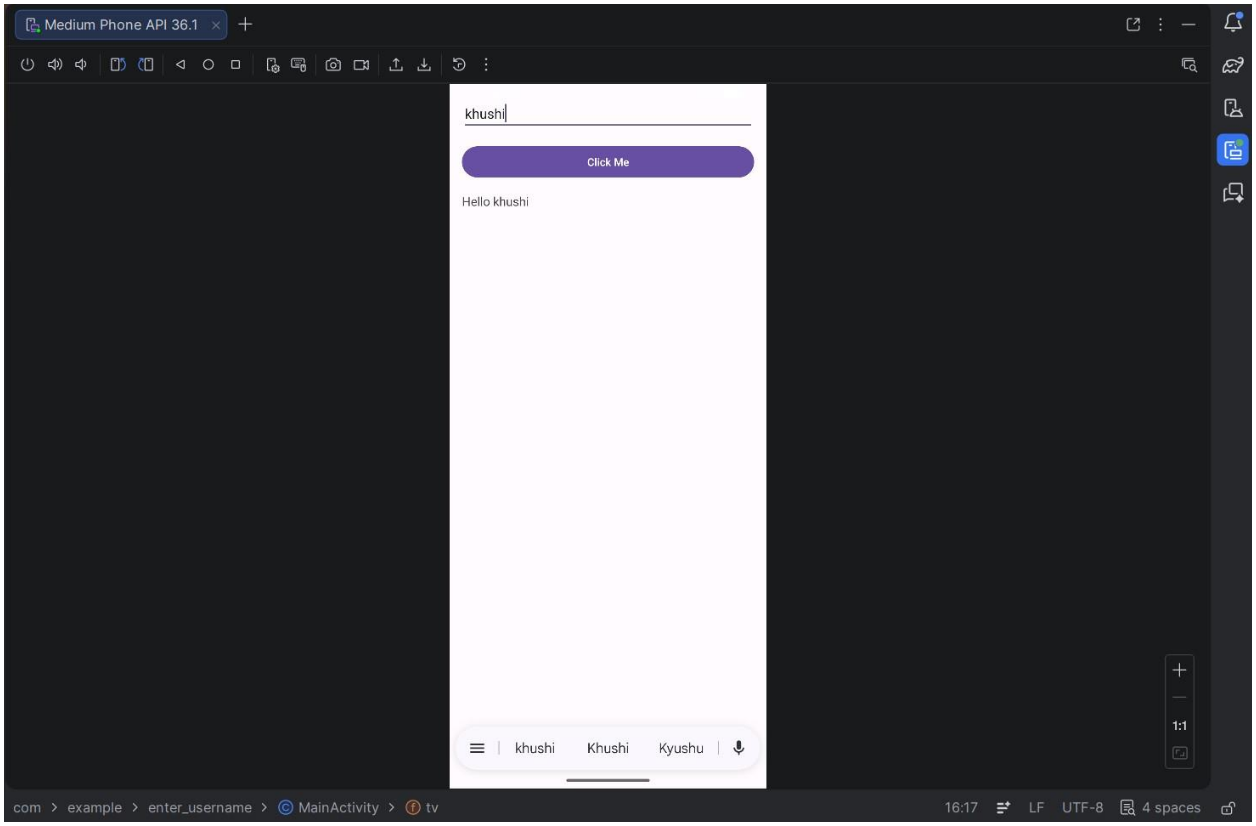
```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"     android:orientation="vertical"
android:padding="16dp">

    <EditText         android:id="@+id/et1"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Enter text here"
android:inputType="text" />

    <Button         android:id="@+id/btn1"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Click Me"
        android:layout_marginTop="16dp" />

    <TextView         android:id="@+id/tv1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Hello Android!"
android:textSize="16sp"
        android:layout_marginTop="16dp" />

</LinearLayout>
```

MainActivity.java:

```java
package com.example.enter_username;
```

```java
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button btn;
    EditText et;
    TextView tv;

    @Override    protected void onCreate(Bundle
savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);

        btn = findViewById(R.id.btn1);
et = findViewById(R.id.et1);
tv = findViewById(R.id.tv1);

        btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                String text = et.getText().toString();

                String message = "Hello " + text;

                tv.setText(message);

                Toast.makeText(MainActivity.this, message, Toast.LENGTH_SHORT).show();
            }
        });
} }
```

Output:



Program 2: Create a screen that has input boxes for User Name, Password,

Address, Gender (radio buttons for male and female), Age (numeric), Date of Birth (Date Picket), State (Spinner) and a Submit button. On clicking the submit button, print all the data below the Submit Button. Use 1. Linear Layout 2. Relative Layout and 3. Grid Layout or Table Layout.

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?<ScrollView
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"     android:layout_height="match_parent">

    <LinearLayout
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:orientation="vertical"
android:padding="16dp">

        <!-- TABLE LAYOUT -->
        <TableLayout
android:layout_width="match_parent"
android:layout_height="wrap_content">

            <!-- Username -->
            <TableRow>
                <TextView android:text="Username:"/>
                <EditText
android:id="@+id/etName"
android:layout_width="match_parent"
android:inputType="text"/>
            </TableRow>

            <!-- Password -->
            <TableRow>
                <TextView android:text="Password:"/>
                <EditText
android:id="@+id/etPassword"
android:layout_width="match_parent"
android:inputType="textPassword"/>
</TableRow>
```

```
            <!-- Address -->
            <TableRow>
                <TextView android:text="Address:"/>
                <EditText
android:id="@+id/etAddress"
android:layout_width="match_parent"
android:inputType="textMultiLine"/>
            </TableRow>

            <!-- Gender -->
            <TableRow>
                <TextView android:text="Gender:"/>
                <RadioGroup
                    android:id="@+id/radioGroup">

                    <RadioButton
android:id="@+id/rbMale"
android:text="Male"/>

                    <RadioButton
android:id="@+id/rbFemale"
android:text="Female"/>
                </RadioGroup>
</TableRow>

            <!-- Age -->
            <TableRow>
                <TextView android:text="Age:"/>
                <EditText
android:id="@+id/etAge"
android:layout_width="match_parent"
android:inputType="number"/>
            </TableRow>

            <!-- DOB -->             <TableRow>
de<TextView android:text="DOB:"/>
                <DatePicker
android:id="@+id/datePicker"
                    android:layout_width="wrap_content"/>
            </TableRow>

            <!-- State -->
            <TableRow>
                <TextView android:text="State:"/>
                <Spinner
android:id="@+id/spinnerState"/>
```

```
            </TableRow>

        </TableLayout>

        <!-- SUBMIT BUTTON -->         <Button             android:id="@+id/btnSubmit"
android:layout_width="match_parent"            android:layout_height="wrap_content"
android:text="Submit"
            android:layout_marginTop="20dp"/>

        <!-- OUTPUT -->         <TextView            android:id="@+id/tvResult"
android:layout_width="match_parent"            android:layout_height="wrap_content"
android:layout_marginTop="20dp"
            android:textSize="16sp"/>

    </LinearLayout>

</ScrollView>
```

MainActivity.java:

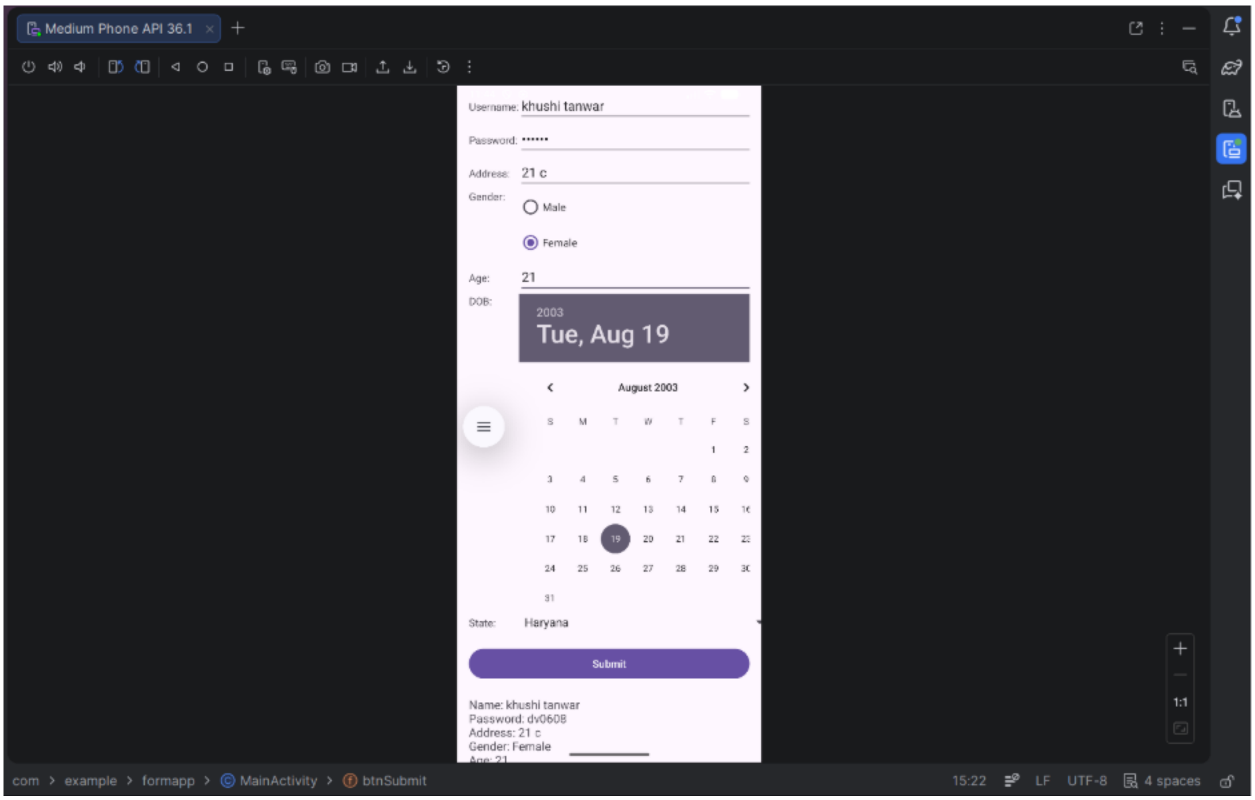
```
package com.example.formapp;

import android.os.Bundle;
import android.view.View;
import android.widget.*;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extend AppCompatActivity {
    EditText etName, etPassword, etAddress, etAge;
    RadioGroup radioGroup;
Spinner spinner;
    DatePicker datePicker;
    Button         btnSubmit;
TextView tvResult;

    String[] states = {"Haryana", "Delhi", "Punjab", "UP", "Rajasthan"};

    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);

        etName = findViewById(R.id.etName);
etPassword = findViewById(R.id.etPassword);
etAddress = findViewById(R.id.etAddress);
etAge = findViewById(R.id.etAge);
radioGroup = findViewById(R.id.radioGroup);
spinner = findViewById(R.id.spinnerState);
datePicker = findViewById(R.id.datePicker);
btnSubmit = findViewById(R.id.btnSubmit);
tvResult = findViewById(R.id.tvResult);

        // Spinner setup
        ArrayAdapter<String> adapter = new ArrayAdapter<>(
this,
android.R.layout.simple_spinner_item,
states
        );
adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);
spinner.setAdapter(adapter);

        btnSubmit.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                String name = etName.getText().toString();
                String pass = etPassword.getText().toString();
                String address = etAddress.getText().toString();
String age = etAge.getText().toString();

                int selectedId = radioGroup.getCheckedRadioButtonId();
                RadioButton rb = findViewById(selectedId);
                String gender = (rb != null) ? rb.getText().toString() : "Not Selected";

                int day = datePicker.getDayOfMonth();
int month = datePicker.getMonth() + 1;
int year = datePicker.getYear();

                String dob = day + "/" + month + "/" + year;

                String state = spinner.getSelectedItem().toString();

                String result =
                        "Name: " + name + "\n" +
```

```
"Password: " + pass + "\n" +
```

```
                                "Address: " + address + "\n" +
                                "Gender: " + gender + "\n" +
                                "Age: " + age + "\n" +
                                "DOB: " + dob + "\n" +
                                "State: " + state;

                tvResult.setText(result);
            }
        });
    }
}
```

Output:



Program 3: Develop an application that inserts some notifications into Notification area and whenever a notification is inserted, it should show a toast with details of the notification.

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"     android:gravity="center"
android:orientation="vertical"     android:padding="20dp">

    <TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Notification App"
android:textSize="22sp"
android:textStyle="bold"
        android:layout_marginBottom="30dp"/>

    <Button
android:id="@+id/btnNotify"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Show Notification"/>

</LinearLayout>
```

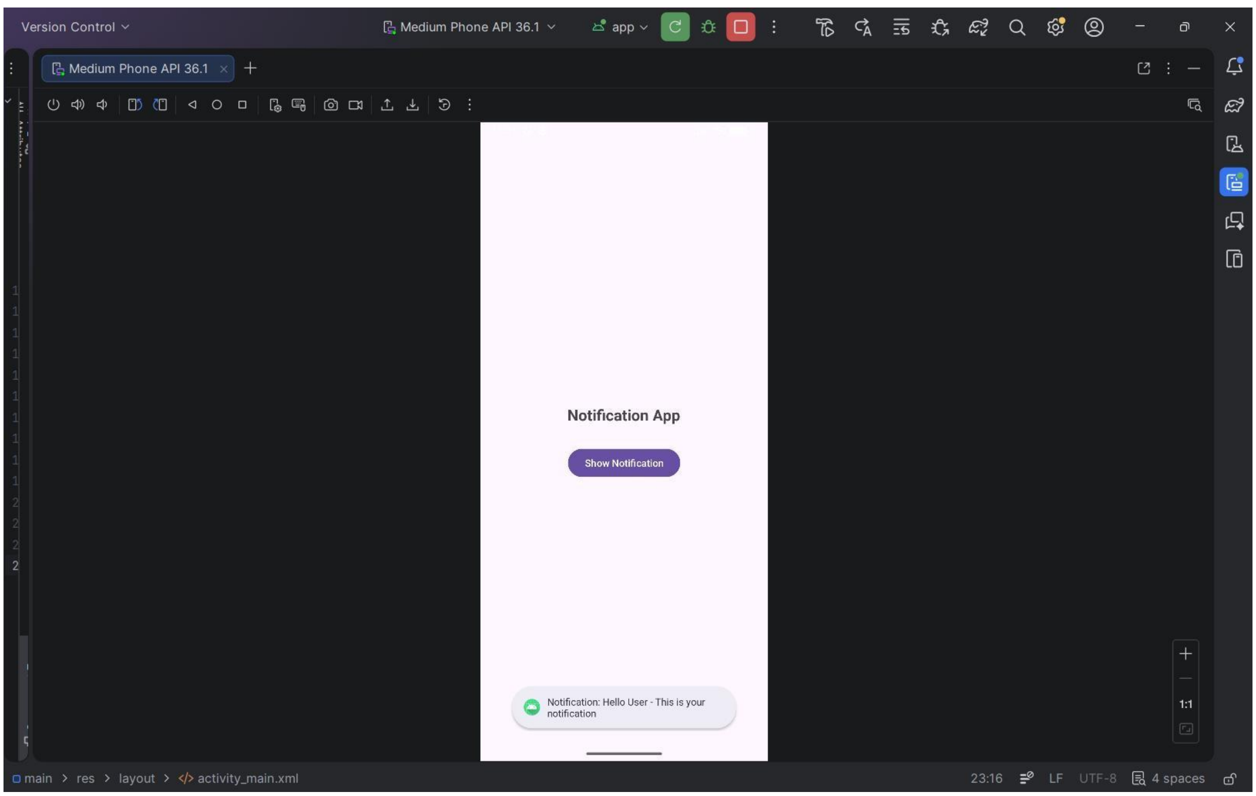
MainActivity.java:

```
package com.example.notificationapp;
```

```java
import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.os.Build; import
android.os.Bundle; import
android.widget.Button;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity; import
androidx.core.app.NotificationCompat;

public class MainActivity extends AppCompatActivity {

    Button btnNotify;
    String CHANNEL_ID = "my_channel";

    @Override     protected void onCreate(Bundle
savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);

        btnNotify = findViewById(R.id.btnNotify);

        btnNotify.setOnClickListener(v -> {

            String title = "Hello User";
            String message = "This is your notification";

            // Toast showing notification details
            Toast.makeText(this,
                    "Notification: " + title + " - " + message,
                    Toast.LENGTH_LONG).show();

            NotificationManager manager =
                    (NotificationManager) getSystemService(NOTIFICATION_SERVICE);

            // For Android 8+
            if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
NotificationChannel channel =                       new
NotificationChannel(                              CHANNEL_ID,
                                "My Channel",
                                NotificationManager.IMPORTANCE_DEFAULT);

                manager.createNotificationChannel(channel);
            }
```

```
            NotificationCompat.Builder builder =
new NotificationCompat.Builder(this, CHANNEL_ID)
                            .setContentTitle(title)
                            .setContentText(message)
                            .setSmallIcon(android.R.drawable.ic_dialog_info)
.setAutoCancel(true);

            manager.notify(1, builder.build());
        });
    }
}
```

Output:



Program 4: Develop an application that uses a menu with 3 options for dialing a number, opening a website and to send an SMS. On selecting an option, the appropriate action should be invoked using intents.

activity_main.xml:

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"     android:gravity="center"
android:orientation="vertical">

    <TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Menu Intent App"
android:textSize="22sp"
        android:textStyle="bold"/>

</LinearLayout>
```

menu_main.xml:

```xml
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">

    <item
android:id="@+id/menu_call"
android:title="Dial Number"/>

    <item
android:id="@+id/menu_web"
        android:title="Open Website"/>

    <item
android:id="@+id/menu_sms"
android:title="Send SMS"/>

</menu>
```

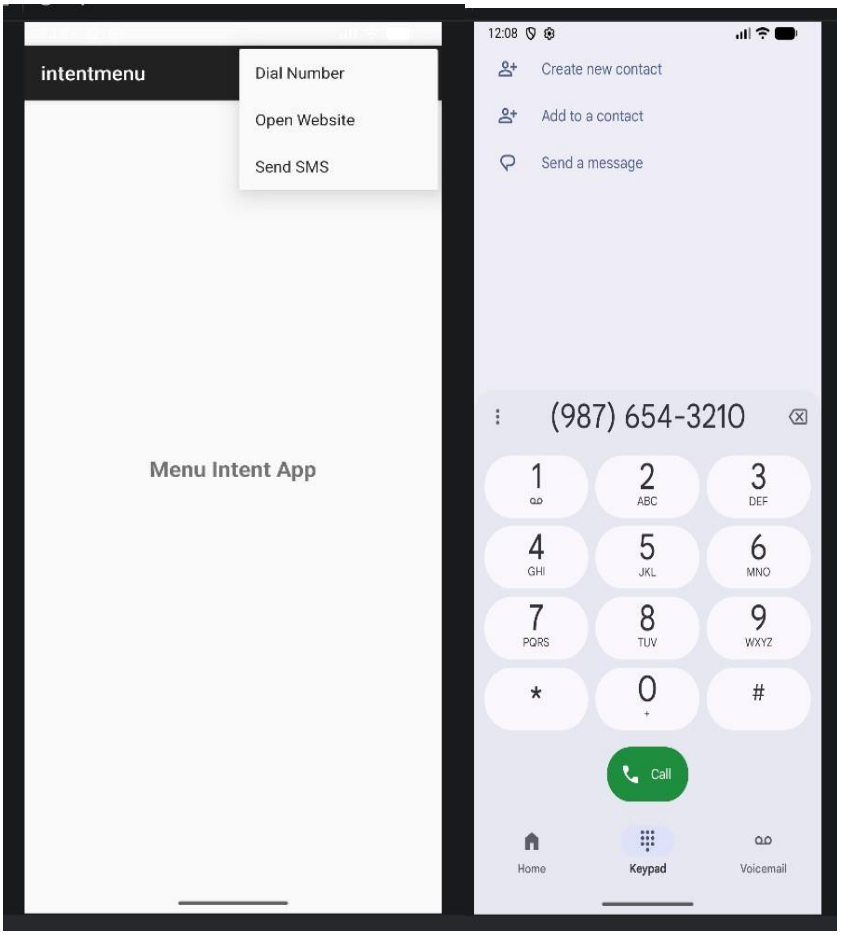
MainActivity.java:

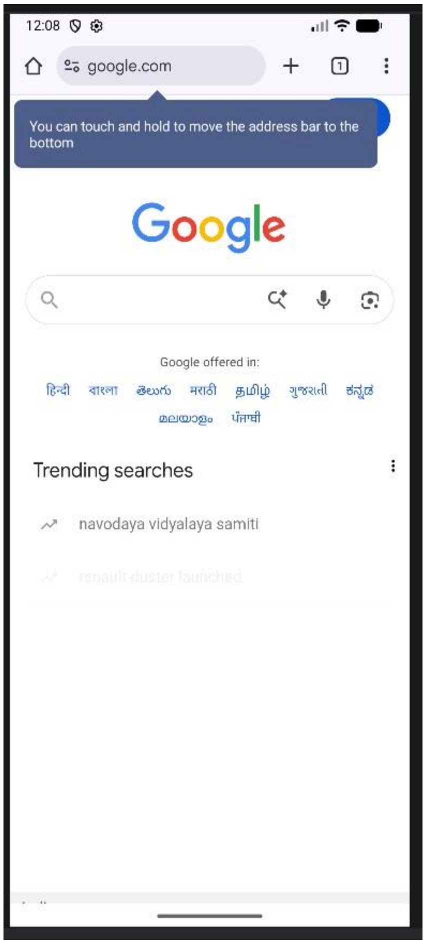
```java
package com.example.intentmenu;

import android.content.Intent; import android.net.Uri;
import android.os.Bundle;
```

```java
import android.view.Menu;
import android.view.MenuItem;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

  @Override     protected void onCreate(Bundle
savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
    }

    // Create menu     @Override     public boolean
onCreateOptionsMenu(Menu menu) {
getMenuInflater().inflate(R.menu.menu_main, menu);
return true;
    }

    // Handle menu clicks
    @Override
    public boolean onOptionsItemSelected(MenuItem item) {

        int id = item.getItemId();

        //  Dial Number
        if (id == R.id.menu_call) {
            Intent intent = new Intent(Intent.ACTION_DIAL);
intent.setData(Uri.parse("tel:9876543210"));
startActivity(intent);            return true;
        }

        // Open Website
        if (id == R.id.menu_web) {
            Intent intent = new Intent(Intent.ACTION_VIEW);
intent.setData(Uri.parse("https://www.google.com"));
startActivity(intent);            return true;
        }

        //  Send SMS
        if (id == R.id.menu_sms) {
            Intent intent = new Intent(Intent.ACTION_VIEW);
intent.setData(Uri.parse("sms:9876543210"));
intent.putExtra("sms_body", "Hello from my app!");
```

```
            startActivity(intent);
return true;
        }

        return super.onOptionsItemSelected(item);
    }
}
```

Output:


intentmenu
Dial Number
Open Website
Send SMS
Menu Intent App
12:08
Create new contact
Add to a contact
Send a message
(987) 654-3210
1
2
ABC
3
DEF
4
GHI
5
JKL
6
MNO
7
PQRS
8
TUV
9
WXYZ
*
0
+
#
Call
Home
Keypad
Voicemail



Program 5: Create an application that uses a text file to store user names and passwords (tab separated fields and one record per line) When the user submits a login name and password through a screen, the details should be verified with the text file data and if they match, show a dialog saying that login is successful. Otherwise, show the dialog with Login Failed message.

activity_main.xml:

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:gravity="center"     android:orientation="vertical"
android:padding="20dp">
```

```
    <TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Login System"
android:textSize="22sp"         android:textStyle="bold"
        android:layout_marginBottom="30dp"/>

    <EditText         android:id="@+id/etUser"
android:layout_width="match_parent"
android:layout_height="wrap_content"
        android:hint="Enter Username"/>

    <EditText         android:id="@+id/etPass"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Enter Password"
android:inputType="textPassword"
        android:layout_marginTop="10dp"/>

    <Button         android:id="@+id/btnLogin"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Login"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

MainActivity.java:

```
package com.example.filelogin;

import android.app.AlertDialog;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;

import androidx.appcompat.app.AppCompatActivity;

import java.io.BufferedReader;
import java.io.InputStreamReader;

public class MainActivity extends AppCompatActivity {
```

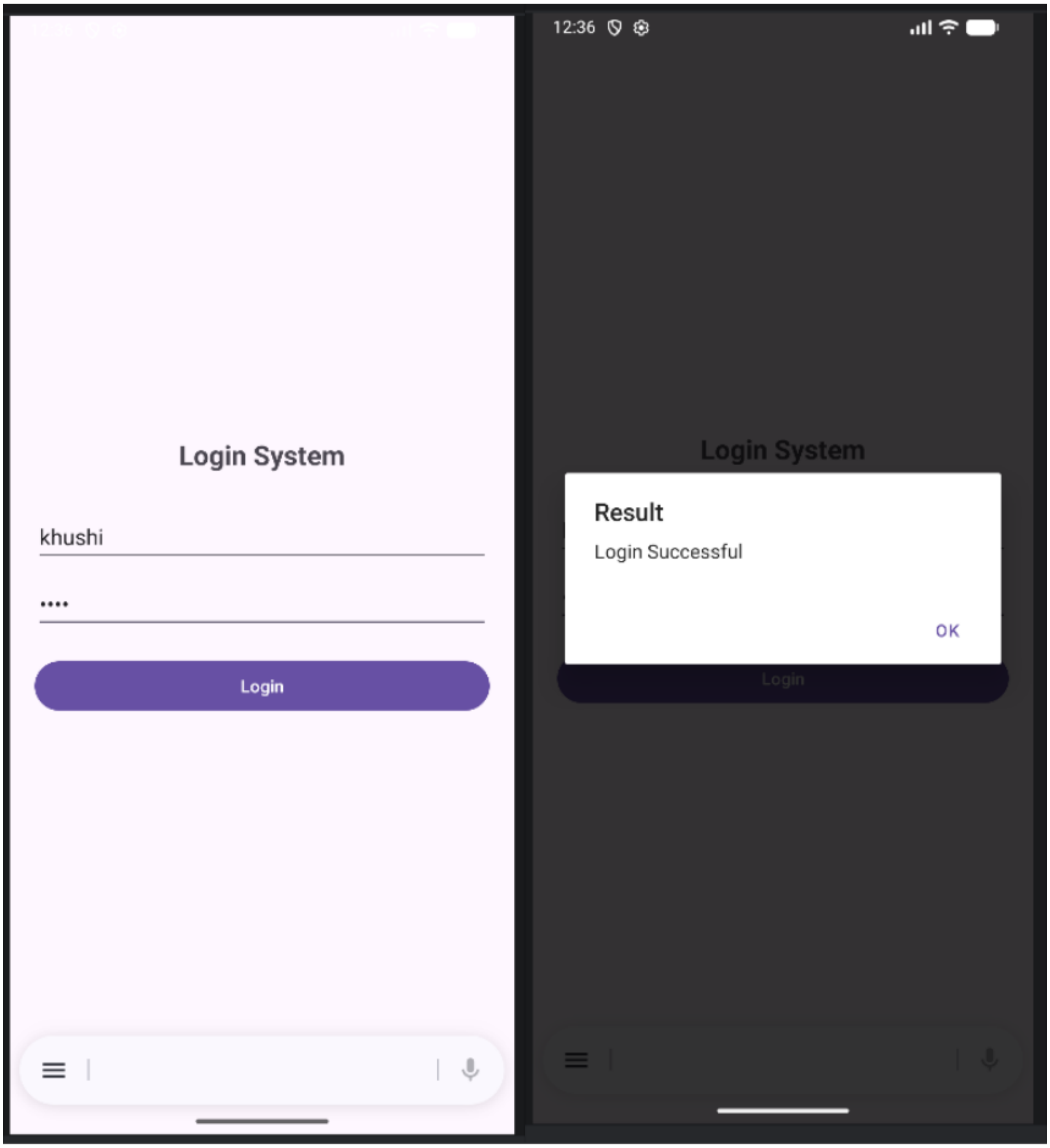
```
    EditText etUser, etPass;
    Button btnLogin;

    @Override     protected void onCreate(Bundle
savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);

        etUser = findViewById(R.id.etUser);
etPass = findViewById(R.id.etPass);
btnLogin = findViewById(R.id.btnLogin);

        btnLogin.setOnClickListener(v -> {

            String inputUser = etUser.getText().toString();
            String inputPass = etPass.getText().toString();

            boolean isValid = false;

            try {
                BufferedReader reader = new BufferedReader(
new InputStreamReader(getAssets().open("users.txt"))                 );

                String line;

                while ((line = reader.readLine()) != null) {

                    String[] parts = line.split("\t");

                    if (parts.length == 2) {
String fileUser = parts[0];
                        String filePass = parts[1];

                        if (inputUser.equals(fileUser) && inputPass.equals(filePass)) {
isValid = true;                            break;
                        }
                    }
}

                reader.close();

            } catch (Exception e) {
                e.printStackTrace();
            }
```

```
            if (isValid) {
showDialog("Login Successful");
            } else {
showDialog("Login Failed");
            }

        });
}

    private void showDialog(String message) {
new AlertDialog.Builder(this)
.setTitle("Result")
                .setMessage(message)
                .setPositiveButton("OK", null)
                .show();
    }
}
```

Users.txt:

```
dev      123
shweta   456
raghav   789
krishna 1011
khushi  1022
```

Output:



Program 6: Create an activity to navigate to different activities through Intents- Implicit and Explicit

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"     android:gravity="center"
android:orientation="vertical"     android:padding="20dp">

    <TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Intent Demo"
android:textSize="22sp"
android:textStyle="bold"
        android:layout_marginBottom="30dp"/>

    <Button
android:id="@+id/btnExplicit"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Go to Second Activity"
android:layout_marginBottom="15dp"/>

    <Button
android:id="@+id/btnWebsite"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Open Website"
        android:layout_marginBottom="15dp"/>

    <Button
android:id="@+id/btnDial"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Dial Number"/>

</LinearLayout>
```

MainActivity.java:

```java
package com.example.intentdemo;

import com.example.intentdemo.SecondActivity;
import android.content.Intent; import
android.net.Uri; import android.os.Bundle;
import android.widget.Button;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button btnExplicit, btnWebsite, btnDial;

    @Override     protected void onCreate(Bundle
savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);

        btnExplicit = findViewById(R.id.btnExplicit);
btnWebsite = findViewById(R.id.btnWebsite);
btnDial = findViewById(R.id.btnDial);

        // ◈ EXPLICIT INTENT
        btnExplicit.setOnClickListener(v -> {
            Intent intent = new Intent(MainActivity.this, SecondActivity.class);
startActivity(intent);
        });

        //     ◈     IMPLICIT     INTENT     -     OPEN     WEBSITE
btnWebsite.setOnClickListener(v -> {
            Intent intent = new Intent(Intent.ACTION_VIEW);
intent.setData(Uri.parse("https://www.google.com"));
startActivity(intent);
        });

        // ◈ IMPLICIT INTENT - DIAL
btnDial.setOnClickListener(v -> {
            Intent intent = new Intent(Intent.ACTION_DIAL);
intent.setData(Uri.parse("tel:9876543210"));
startActivity(intent);
        });
    }
}
```

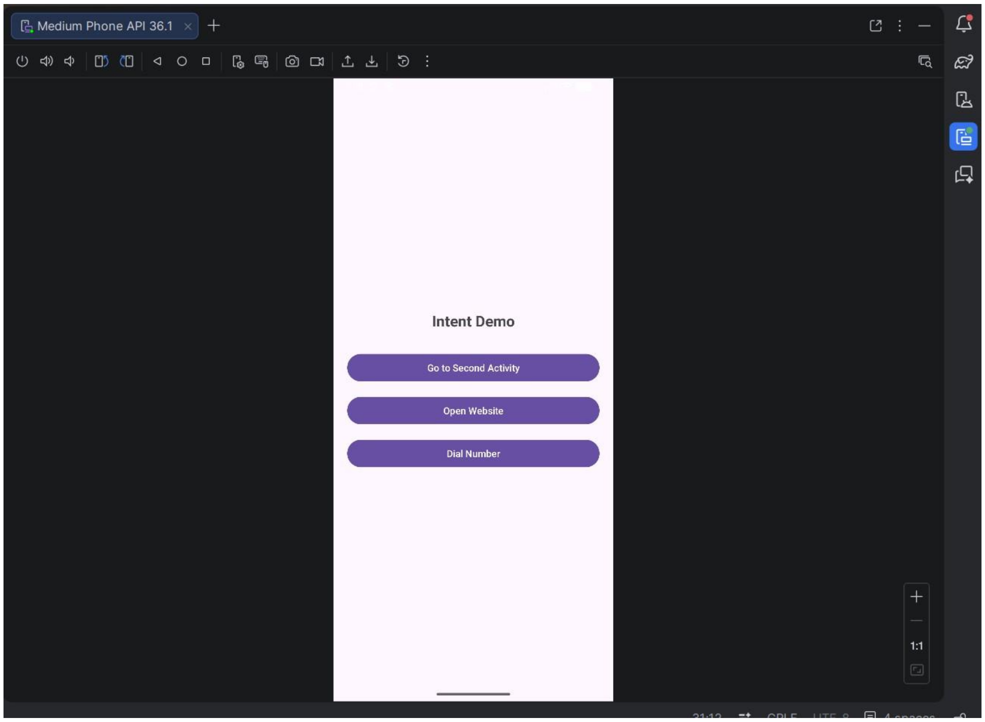
activity_second.xml:

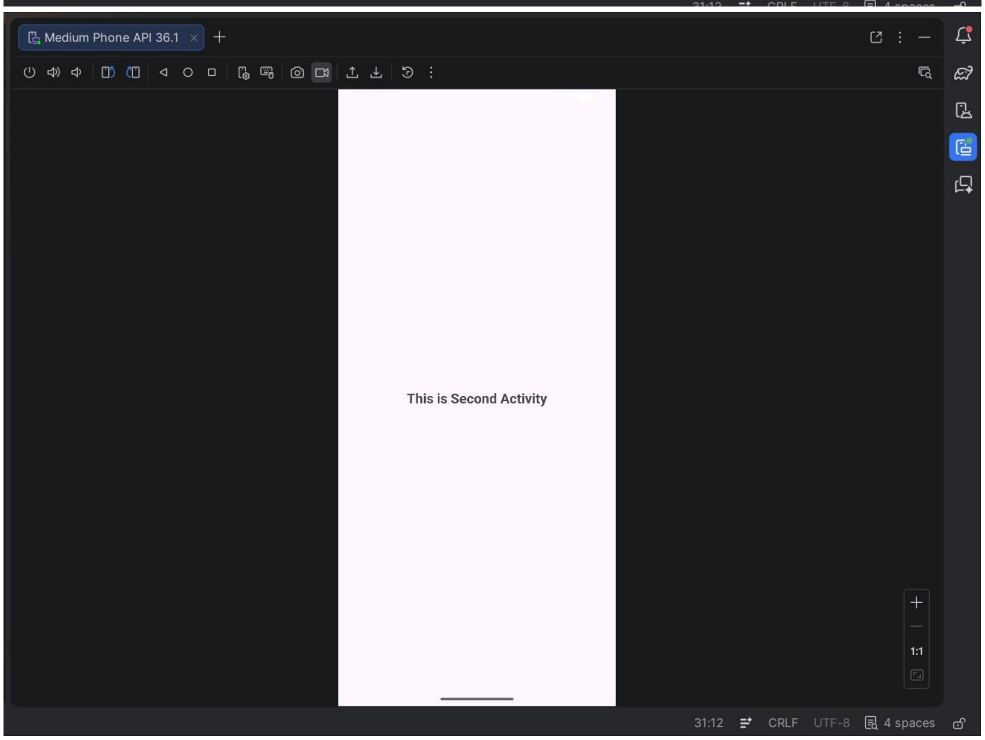
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"     android:gravity="center"
android:orientation="vertical">

    <TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="This is Second Activity"
android:textSize="20sp"
        android:textStyle="bold"/>

</LinearLayout>
```

SecondActivity.java:

```
package com.example.intentdemo;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class SecondActivity extends AppCompatActivity {

    @Override     protected void onCreate(Bundle
savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_second);
    }
}
```

Output:


Medium Phone API 36.1
Intent Demo
Go to Second Activity
Open Website
Dial Number
1:1


Medium Phone API 36.1
This is Second Activity
1:1
31:12
CRLF
UTF-8
4 spaces

Program 7: Create an activity to create option Menu.

activity_main.xml:

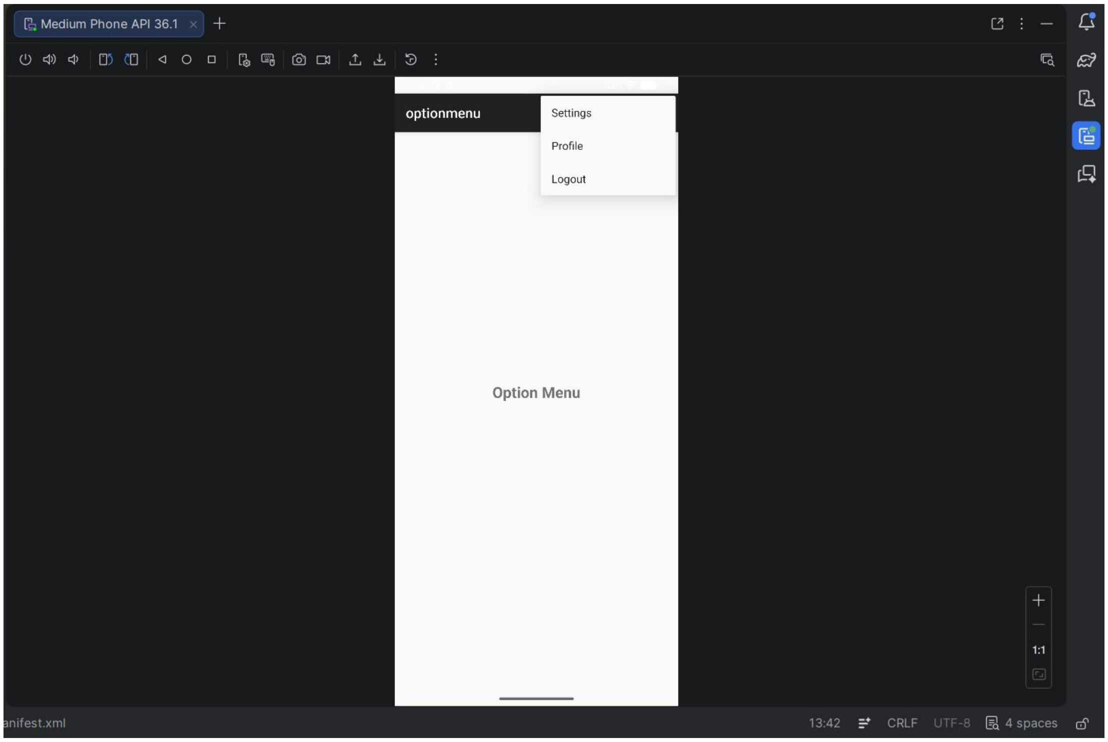
```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"     tools:context=".MainActivity">

    <TextView
android:id="@+id/textView"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Option Menu"
android:textSize="22sp"
android:textStyle="bold"
        android:layout_centerInParent="true" />

</RelativeLayout>
```

MainActivity.java:

```
package com.example.optionmenu;

import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override     protected void onCreate(Bundle
savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
    }

    // Step 1: Create Option Menu
    @Override     public boolean
onCreateOptionsMenu(Menu menu) {
menu.add(0, 1, 0, "Settings");         menu.add(0,
2, 0, "Profile");
```

```
        menu.add(0, 3, 0, "Logout");
return true;
    }

    // Step 2: Handle Menu Clicks
    @Override
    public boolean onOptionsItemSelected(MenuItem item) {

        switch (item.getItemId()) {
case 1:
                Toast.makeText(this,   "Settings   Clicked",   Toast.LENGTH_SHORT).show();
return true;

            case 2:
                Toast.makeText(this,   "Profile   Clicked",   Toast.LENGTH_SHORT).show();
return true;

            case 3:
                Toast.makeText(this,   "Logout   Clicked",   Toast.LENGTH_SHORT).show();
return true;

            default:
                return super.onOptionsItemSelected(item);
}
    } }
```

Output:



Program 8: Create an activity using Button Controls- CheckBox, RadioButtons, Radio Groups

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"     android:layout_height="match_parent"
android:orientation="vertical"
    android:padding="20dp">

    <!-- CheckBox -->
    <CheckBox
```

```
        android:id="@+id/checkBox"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
        android:text="Accept Terms and Conditions" />

    <!-- RadioGroup -->
    <RadioGroup
android:id="@+id/radioGroup"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="20dp">

        <RadioButton
android:id="@+id/radioMale"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Male" />

        <RadioButton
android:id="@+id/radioFemale"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Female" />

    </RadioGroup>

    <!-- Button -->
    <Button
android:id="@+id/btnSubmit"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Submit"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

MainActivity.java:

```
package com.example.buttoncontrols;


import android.os.Bundle; import android.view.View;
import android.widget.Button; import
android.widget.CheckBox; import
android.widget.RadioButton; import
android.widget.RadioGroup; import android.widget.Toast;
```

```java
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    CheckBox checkBox;
    RadioGroup radioGroup;
Button btnSubmit;

    @Override     protected void onCreate(Bundle
savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);

        checkBox = findViewById(R.id.checkBox);
radioGroup = findViewById(R.id.radioGroup);
btnSubmit = findViewById(R.id.btnSubmit);

        btnSubmit.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                // CheckBox state
                boolean isChecked = checkBox.isChecked();

                // RadioButton selected
                int selectedId = radioGroup.getCheckedRadioButtonId();

                if (selectedId == -1) {
                    Toast.makeText(MainActivity.this, "Select Gender",
Toast.LENGTH_SHORT).show();                     return;
                }

                RadioButton   selectedRadio   =   findViewById(selectedId);
String gender = selectedRadio.getText().toString();

                // Show result
                String message = "Accepted: " + isChecked + "\nGender: " + gender;

                Toast.makeText(MainActivity.this, message, Toast.LENGTH_LONG).show();
            }
        });
    }
}
```

Output: